

# Vignesh S

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## PROFESSIONAL SUMMARY

Full-stack software engineer with 15+ months building production IoT, SaaS, and e-commerce platforms end-to-end — from MongoDB schema design and secure payment gateway integration to real-time dashboards and CMS-driven storefronts. Strong backend instincts: multi-tenant architecture, background workers, webhook handling, and idempotent payment flows.

## EXPERIENCE

### Software Engineer

Mar 2025 – Present

*AAPGS Private Limited*

*Madurai, Tamilnadu*

- Reduced database write operations by 98% by buffering IoT payloads in Redis and flushing via background worker using `insertMany`, replacing per-event individual inserts.
- Built REST APIs and Angular/React UIs for IoT and SaaS products, owning features end-to-end from MongoDB schema design through frontend integration.
- Designed multi-tenant architecture with dynamic per-tenant database switching and JWT-based authentication.
- Engineered OTP-based delivery verification for e-commerce platform, generating OTP via UPI request to PayU pre-delivery and validating against delivery agent and customer input to prevent fraudulent order confirmation.
- Built idempotent PayU payment integration with hash-based signature verification and retry-safe `txn.id` handling to prevent duplicate transactions.

### Web Developer

Dec 2024 – Feb 2025

*E-Careerpluz Private Limited*

*Madurai, Tamilnadu*

- Built a responsive business website with PHP backend handling contact forms, subscription emails, and notifications.
- Crafted interactive UI using HTML, CSS, and JavaScript for a digital services and legal advisory platform.

## PROJECTS

### DeployDock | *Next.js, Node.js, TypeScript, MongoDB, SSH, PM2, SSE* | [GitHub](#)

Personal Project

- Built a self-hosted CI/CD platform with SSH-based remote deployments, PM2 process management, and git-based rollback support.
- Designed multi-tenant isolation using per-tenant MongoDB databases and AES-256-CBC encrypted SSH credentials.
- Streamed deploy logs in real time via SSE backed by worker threads, keeping the deployment pipeline non-blocking under concurrent jobs.
- Automated deployments with GitHub OAuth and webhook integration, tracking commit SHAs and full deployment history.
- Supported Docker-based deployments alongside SSH, allowing containerized apps to be deployed and managed through the same CI/CD pipeline.

### ProfluxIoT | *Angular, Node.js, MongoDB, Redis, Socket.IO, ECharts*

AAPGS Private Limited

- Architected a scalable IoT platform with per-tenant database isolation and JWT-based authentication.
- Delivered a configurable dashboard system with drag-and-drop real-time ECharts and table widgets with user-level assignment.
- Eliminated per-emit MongoDB reads by caching dashboard metadata in Redis with NX locks, reducing query latency on every real-time data push.
- Engineered a visual automation engine using FoblexFlow enabling conditional workflows (HTTP, Email, WhatsApp) per device.

## TECHNICAL SKILLS

**Languages:** JavaScript, TypeScript, Python

**Frameworks:** React, Next.js, Angular, Node.js, Express.js

**Databases:** MongoDB, Redis

**Tools & Libraries:** Socket.IO, PayU, Git, PM2, GrapesJS

## EDUCATION

### The Madura College

*Bachelor of Science in Information Technology, CGPA: 7.61*

Madurai, Tamilnadu

*Aug 2021 – May 2024*